



**THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MBINGA DISTRICT COUNCIL
FORM IV DISTRICT PRE MOCK EXAMINATIONS MARCH 2026**



032

CHEMISTRY

Time: 3:00 HOURS

Monday 23rd March 2026 A.m.

INSTRUCTIONS

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1. This paper consists of section A, B and C with a total of **eleven (11)** questions
2. Answer **all** questions in section A and B and **two (2)** questions from section C
3. Section A carries **sixteen (16)** marks, section B **fifty-four (54)** marks and section C carries **thirty (30) marks**.
4. All writing should be in Blue or Black pen **except** for the diagrams that must be drawn in pencil
5. Communication devices and any unauthorized materials are not allowed in the examination room
6. Write your name on every page of your answer sheet provided.
7. Where necessary the following constants may be used;
Atomic masses H=1, C=12, N=14, O=16, Na=23, Ca=40, Cl=35.5, Cu=64, Zn=65
Avogadro's number = 6.02×10^{23}
GMV at STP = 22.4 dm^3 1litre = $1 \text{ dm}^3 = 1000 \text{ cm}^3$

Section A: (16 Marks)

1. For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided.
 - i. The following are elements which do not react with chlorine
A. argon B. sodium C. potassium D. sulphur E. beryllium
 - ii. One among the following substances is made by application of chemistry in agriculture: A. Yeast B. petrol C. Toothpaste D. Pesticides E. Alcohols
 - iii. A form one student dissolved one mole salts M into water and the solution was tested by litmus paper. The salt was found to change blue litmus paper red. This salt is likely to be;
A. Normal B. Complex C. Basic D. Acidic E. Neutral
 - iv. Loose or floppy clothing is not allowed in the laboratory. Why?
A. Movement has to be fast B. It will get wet when water splashes C. It may catch fire or cause one to fall D. It causes poor ventilation in the body E. It prevents experiment from being conducted well
 - v. Which of the following is not true about bases? A. Most bases have bitter taste B. Bases are slippery when rubbed between fingers C. Bases neutralize acids to form

- salt and water only D. Weak base is the one that dissociate completely in solution
E. Are compounds which have oxide or hydroxide
- vi. A simple proof that some chemical reactions take place in our bodies is that A. We eat a balanced diet B. Doctors tell us so in the hospitals C. We occasionally fall sick D. The food we eat or the drinks we take are quite different from the waste products from our bodies E. There is no proof
- vii. which of the following is not the components of first aid kit
A. pain kill B. gloves C. dissector D. cotton wool
- viii. Which gas caused acidic rain when combine with rain water?
A. oxygen B. Carbon dioxide C. water D. noble gases
- ix. The welders prefer the use of non- luminous flame for their work simply because _____
A. Produce very hot flame B. It is available C. It is easy to transport
D. Can be made by kerosene
- x. Experts from Kamanga Hospital were doing scientific research about Cholera in Misungwi District Council and obtained the results from their experiment that do not support their hypotheses. What advice could you give to them?
A. The experiment should be changed B. The results should be left out
C. Ideas for further testing to find a solution should be given D. A new problem should be identified E. The observations should be altered
2. Match the materials in List A with the correct method of preventing it from rusting in List B by writing the letter of the correct in answer sheet provided

LIST A	LIST B
i. A colourless gas which is slightly soluble in water, neutral to litmus paper but used as rocket fuel	A. Oxygen
ii. It supports plant survive but less important to combustion	B. Nitrogen
iii. A greenish gas, denser than air and used as military weapon since it is poisonous	C. Sulphur dioxide
iv. It is colourless acidic gas with pungent choking smell, used to control pH in chemical process	D. Hydrogen chloride
v. A non-poisonous gas physically isolated from air and used to make fertilizer	E. carbon dioxide
vi. It has strong choking smell of urine	F. Fluoride
	G. Ammonia
	H. Hydrogen
	I. chlorin

SECTION B (54 Marks)

Answer **all** questions from this section

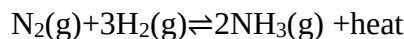
3. a) What is the aim of having
- fume chamber in the chemistry laboratory?
 - Why laboratory doors are open outward?
- b) Briefly give one reason for the following uses of metals
- Aluminum is used for making window frame.
 - Iron is used for making bridges.

- iii. Copper for electrical wiring.
 - iv. Lead for roofing.
 - v. Zinc for coating of iron.
4. A certain compound Q was heated but it didn't decompose. But when Q reacted with dilute hydrochloric acid gas P was produced, the gas P turned lime water milky. When compound Q was burned on a direct flame it burned with golden-yellow flame.
- a) Identify the compound Q and gas P
 - b) Write the balanced chemical equation for the reaction between Q and dilute hydrochloric acid.
5. Explain how to handle chemicals having the warning signs of flammable, corrosive, and toxic in the laboratory.
6. The table below shows two brands of bottled water for drinking and the concentration of different mineral ions in each brand. Study the table and answer the questions below it.

Composition in mg/litre	MINERAL	Na ⁺	Ca ²⁺	Mg ²⁺	Cl ⁻	NO ₃ ⁻	SO ₄ ²⁻	Fe ²⁺	F ⁻
	UHA1	42	5	6	17	0.8	10	0	1.8
	AFYA	30	3	1.5	8	1	7	0	0.5

- i. Which brand of water is more hard: Explain
 - ii. With aid of chemical equation, How the brands of water mentioned in (i) above can be soften?
 - iii. State the benefit of Taking Calcium contained water
 - iv. Tap water is usual treated before being used. State what is added to perform that function
 - v. Explain one economic effect of using hard water
7. a) The rate of chemical reaction can be affected by different factors. Explain any three factors that affect the rate of chemical reaction.
- b) Sulphuric acid act as dehydrating agent, oxidizing agent as well as an acid. Decide by a very brief answer how it behaving in each of the following cases.
- i. It evolves hydrogen gas when reacting with magnesium.
 - ii. It is used to dry hydrogen chloride gas.
 - iii. It reacts with sodium hydroxide to form salt and water.
 - iv. It releases Sulphur dioxide gas when reacting with copper.
 - v. It turns sugar black
 - vi. It turns copper sulphate white
8. (a) A form three student performed two different experiments
- Experiment I; Students dissolved one tea spoon of sugar in cold water
- Experiment II: Students dissolved one tea spoon of sugar in hot water
- By using the concept of collision theory explain in which experiment the sugar dissolved more quickly?

(b) You are provided with the following equilibrium reaction



What happens to the production of ammonia if

- (i) Pressure increased
- (ii) The equilibrium system is cooled
- (iii) Hydrogen is removed
- (iv) Nitrogen increased

Section C: (30 Marks)

Answer Only Two Questions

9. A form four student had a task to mop the laboratory chamber, accidentally bees attacked and bite on her hand. The technician advised to apply ashes powder to the bite surface. Explain five importance of the process taking place on the skin surface to our daily life after applying the ashes powder.
10. a) How can you tell water is polluted? Give two ways
b) We have coal at Ruanda in Ruvuma region. Authorities in the government have allowed use of coal for domestic and industrial purpose. What warning can you raise concerning likely effects? Give five points.
11. (a) State first Faraday law of electrolysis and second Faraday law of electrolysis
(b) Name three factors which will determine the production of substance released in the electrodes during electrolysis.
(c) What are three applications of electrolysis in daily life?
(d) A steel spoon was electroplated with silver. How long should a current of 1.5A be passed through silver nitrate solution in order to deposit 54 g of silver metal on the spoon